

TimescaleDB vs PostgreSQL Performance Dashboard

 **Total Records**

20

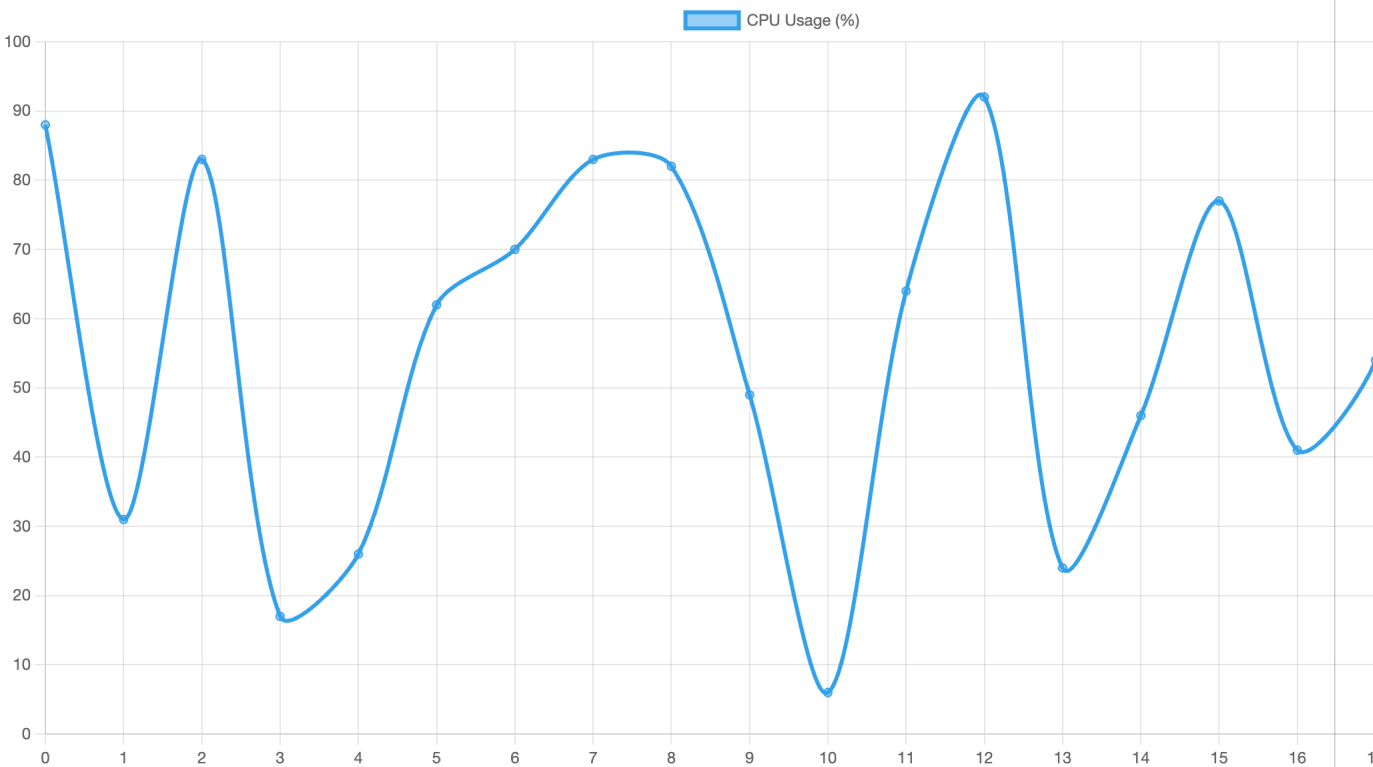
 **PostgreSQL**

Normal Table Performance

 **TimescaleDB**

Hypertable Optimized Performance

CPU Usage Trend



PostgreSQL vs TimescaleDB Query Performance

 **PostgreSQL**

Standard PostgreSQL table using normal time-series storage.

Execution Plan

HashAggregate (cost=54612.48..79378.25 rows=1000006 width=40) (actual time=310.196..310.362 rows=279 loops=1)

Group Key: date_trunc('hour'::text, "time")

Planned Partitions: 64 Batches: 1 Memory Usage: 849kB

 **TimescaleDB**

Hypertable optimized for time-series workloads.

Execution Plan

Finalize HashAggregate (cost=16967.78..16970.78 rows=200 width=16) (actual time=181.779..184.645 rows=279 loops=1)

Group Key: (time_bucket('01:00:00'::interval, system_metrics."time"))

Batches: 1 Memory Usage: 109kB

-> Seq Scan on system_metrics_normal
(cost=0.00..18870.08 rows=1000006 width=12) (actual
time=0.081..182.136 rows=1000006 loops=1)

Planning Time: 12.881 ms

Execution Time: 311.994 ms

-> Gather (cost=8309.54..16964.36 rows=684 width=40)
(actual time=121.697..184.356 rows=582 loops=1)

Workers Planned: 2

Workers Launched: 2

-> Parallel Append (cost=7309.54..15895.96 rows=342
width=40) (actual time=128.468..168.281 rows=194 loops=3)

-> Partial HashAggregate (cost=8579.71..8582.21 rows=200
width=40) (actual time=94.989..95.024 rows=123 loops=3)

Group Key: time_bucket('01:00:00'::interval,
_hyper_1_1_chunk."time")

Batches: 1 Memory Usage: 64kB

Worker 0: Batches: 1 Memory Usage: 64kB

Worker 1: Batches: 1 Memory Usage: 64kB

-> Parallel Seq Scan on _hyper_1_1_chunk
(cost=0.00..7359.22 rows=244098 width=16) (actual
time=0.050..64.313 rows=195278 loops=3)

-> Partial HashAggregate (cost=7309.54..7312.04 rows=200
width=40) (actual time=109.848..109.862 rows=107 loops=2)

Group Key: time_bucket('01:00:00'::interval,
_hyper_1_3_chunk."time")

Batches: 1 Memory Usage: 64kB

Worker 1: Batches: 1 Memory Usage: 64kB

-> Parallel Seq Scan on _hyper_1_3_chunk
(cost=0.00..6091.38 rows=243631 width=16) (actual
time=0.024..68.294 rows=207086 loops=2)

Planning Time: 17.240 ms

Execution Time: 187.089 ms

 Live System Metrics

Time	CPU Usage (%)	Category
0	88	System
1	31	System
2	83	System
3	17	System
4	26	System
5	62	System
6	70	System
7	83	System
8	82	System
9	49	System
10	6	System
11	64	System
12	92	System
13	24	System
14	46	System
15	77	System
16	41	System
17	54	System
18	100	System
19	6	System